

SARAH TISE

stise2@uic.edu | sarahtise101@gmail.com | +1 (630)-267-4730 | Hoffman Estates, IL

EDUCATION

University of Illinois Chicago | Chicago, IL

- Bachelor of Science in Mechanical Engineering - Current GPA: 3.13 (Expected Graduation Spring 2027)

WORK EXPERIENCE

EH Wachs - ITW Division | Lincolnshire, IL

Manufacturing Engineering Intern | May 2026 - August 2026

- Co-Leader of Automated Manufacturing Work Instructions program design & implementation
- Manufacturing process analysis & improvement

Orthopedics Biomechanics Lab at UIC | Chicago, IL

Undergraduate Research Assistant | January 2026 - May 2026

- Simulated loads on various specimen-based models of the spine
- Modeled porcine & human scale spines based on real specimen data

TTX | Chicago, IL

Maintenance & Modification Engineering Intern | May 2025 - August 2025

- Drafted models, drawings, assemblies (AutoCAD & SolidWorks)
- Data optimization & automation
- Time Studies on Repair Procedure

OPIRA Lab at UIC | Chicago, IL

Machine Learning for Skin Melanoma Diagnosis Intern / Research Assistant | June 2024 - August 2024

- Machine learning applications to skin cancer diagnosis research & Optical Coherence Tomography (OCT)
- Performed peer review
- CAD various parts of devices/equipment used in lab

PROJECTS / LEADERSHIP

AIAA Student Organization | Chicago, IL

Vice President (Apr 2025 - Aug 2026); Interim Secretary (Aug 2026 - September 2026)

- Developing a liquid-engine rocket vehicle on the Chicago Sphynx Flight Dynamics Subteam
- Facilitating recruitment initiatives and guiding the next generation of team leaders
- Led/Co-led various team projects: Rocket & drone design and fabrication, launches/flights, composite material fin design research and fabrication

NASA University Student Launch Initiative (USLI) 2025-2026

Vehicle Team Lead, Safety Officer, Propulsions Team Co-lead

- Led Launch Vehicle Design & fabrication process
- Designed & implemented hazard mitigation, performed FMEA
- Manufactured a high-powered solid propellant motor under team mentor guidance & supported design with flight simulations

UIC Mechanical Engineering & Design Graphics Final Competition, 2023

- Ping Pong Ball Sorting Device Competition Winner (won most balls sorted in a set time constraint)

SKILLS

Software: SolidWorks, AutoCAD, Arduino, Santos Basic, OpenRocket, RASAero II, PowerPoint, Ansys, Onshape, Matlab, proficient in Microsoft Office Suite

Coding & Data Optimization: Java, C++, Excel, Python, Linux

Hands-On Skills: Optical imaging, basic woodworking, basic machine operating, additive manufacturing, research & lab experience, hand tool proficiency, welding, soldering, basic circuitry/wiring

Methodologies & Frameworks: FMEA, proficiency in ergonomic design principles, industrial design, design for manufacturability, product line simplification in high volume & low volume production environments, time studies, standard operating procedures & standardized work, lean manufacturing, GD&T

Soft Skills: Public speaking, customer communication & managing multiple projects, time management, collaboration, leadership, cross-functional communication

CERTIFICATIONS, TRAINING & LANGUAGES

- Machine Workshop Safety
- Seminar on Sustainable Engineering
- Level 1 High Power Rocketry (HPR)
- Languages: English, Romanian